Spherical-shell Stokes flow benchmark using Underworld3

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Abstract. We reproduce the analytical spherical-shell Stokes benchmark of Thieulot (2017) using Underworld3 (UW3). The benchmark describes three-dimensional incompressible viscous flow in a spherical shell with tangential velocity on both radial boundaries, radial gravity, and a power-law radial viscosity. We consider the two reference cases used by Thieulot (2017): $m = -1$, for which the viscosity is constant, and $m = 3$, for which the viscosity varies by a factor of 16 across the shell. The UW3 calculations use a P_2P_1 Taylor–Hood finite-element discretisation on successively refined meshes with nominal cell size $h = 1/8, \ldots, 1/128$. Velocity and pressure are evaluated using relative volume L_2-norm errors, and radial normal stress is evaluated using relative boundary L_2-norm errors on the inner and outer spherical surfaces. The calculations recover the expected asymptotic behaviour for smooth Stokes solutions, with approximately third-order velocity convergence and second-order pressure convergence. Boundary radial normal-stress and boundary pressure errors also approach second-order convergence at the finest resolutions. These results verify the UW3 curved-domain Stokes discretisation and provide a compact spherical-shell test for velocity, pressure, and boundary-stress post-processing.

1 Introduction

Incompressible Stokes flow in spherical shells is a core numerical problem in global mantle-convection and planetary-interior modelling. Verification in this geometry is more demanding than in Cartesian boxes because the discretisation must handle curved boundaries, a pressure nullspace, spherical-coordinate forcing, radial viscosity variation, and stress quantities evaluated on non-planar surfaces.

Analytical benchmarks are useful because they separate discretisation and implementation errors from differences between numerical codes. The benchmark of Thieulot (2017) provides a closed-form solution for viscous incompressible flow in a spherical shell. The solution is axisymmetric, has no normal flow on the inner and outer boundaries, uses a radial power-law viscosity, and supplies analytical expressions for density, velocity, pressure, body force, and the stress tensor. It therefore tests the Stokes solve and the post-processing of stress-derived boundary quantities. Related analytical solutions and finite-element verification studies in curved shell geometries have been presented by Kramer et al. (2021) and Davies et al. (2022), while boundary normal stress remains a standard diagnostic in spherical geodynamics benchmarks (Zhong et al., 1993, 2008; Liu and King, 2019).

Here we convert the existing UW3 spherical benchmark report into a concise Copernicus-style article. The aim is to document the governing equations, the analytical benchmark structure, the numerical setup, the error metrics, and the observed convergence of velocity, pressure, and boundary stress using the existing UW3 figures and tables.

2 Governing equations and benchmark solution

The benchmark solves the steady incompressible Stokes equations

$$\nabla \cdot \mathbf{u} = 0, \tag{1a}$$

$$\nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{g} = \mathbf{0}, \tag{1b}$$

where $\mathbf{u}$ is velocity, ρ is density, and $\mathbf{g} = -\mathbf{e}_r$ is radial gravity of unit magnitude. The stress is

$$\boldsymbol{\sigma} = -p\mathbf{I} + 2\mu(r)\boldsymbol{\varepsilon}(\mathbf{u}), \tag{2}$$

with pressure p, viscosity μ, and strain-rate tensor

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}\left(\nabla \mathbf{u} + \nabla \mathbf{u}^T\right). \tag{3}$$

Following Thieulot (2017), spherical coordinates (r, θ, ϕ) are used, with θ the polar angle and ϕ the azimuth. All fields are independent of ϕ. The tangential velocity components are prescribed in the separable form

$$u_\theta(r,\theta) = f(r)\sin\theta, \tag{4a}$$

$$u_\phi(r,\theta) = f(r)\sin\theta, \tag{4b}$$

and incompressibility gives the radial velocity

$$u_r(r,\theta) = g(r)\cos\theta, \qquad g(r) = -\frac{2}{r^2}\int r f(r)\, dr. \tag{5}$$

The boundary condition $u_r = 0$ is imposed on the inner and outer shell surfaces $r = R_1$ and $r = R_2$, so the flow is tangential on both boundaries. The viscosity is

$$\mu(r) = \mu_0 r^{m+1}, \tag{6}$$

where $m = -1$ gives constant viscosity and $m = 3$ gives the radially varying case considered here.

The pressure has the form

$$p(r,\theta) = h(r)\cos\theta, \tag{7}$$

where the radial pressure coefficient is

$$h(r) = \frac{2\mu_0}{r} g(r) \quad \text{for } m = -1, \qquad h(r) = \frac{m+3}{r}\mu(r) g(r) \quad \text{for } m \neq -1. \tag{8}$$

For $m \neq -1$, the radial velocity function is obtained from

$$f(r) = \alpha r^{-(m+3)} + \beta r, \tag{9}$$

with α, β, and the integration constant in $g(r)$ chosen so that $g(R_1) = g(R_2) = 0$. The $m = -1$ case has the corresponding logarithmic radial function given by Thieulot (2017). In both cases the density follows from the radial Stokes balance after substituting the analytical velocity, pressure, viscosity, and gravity fields. This construction yields the closed-form density, velocity, pressure, body force, and stress tensor used for the UW3 error calculations.

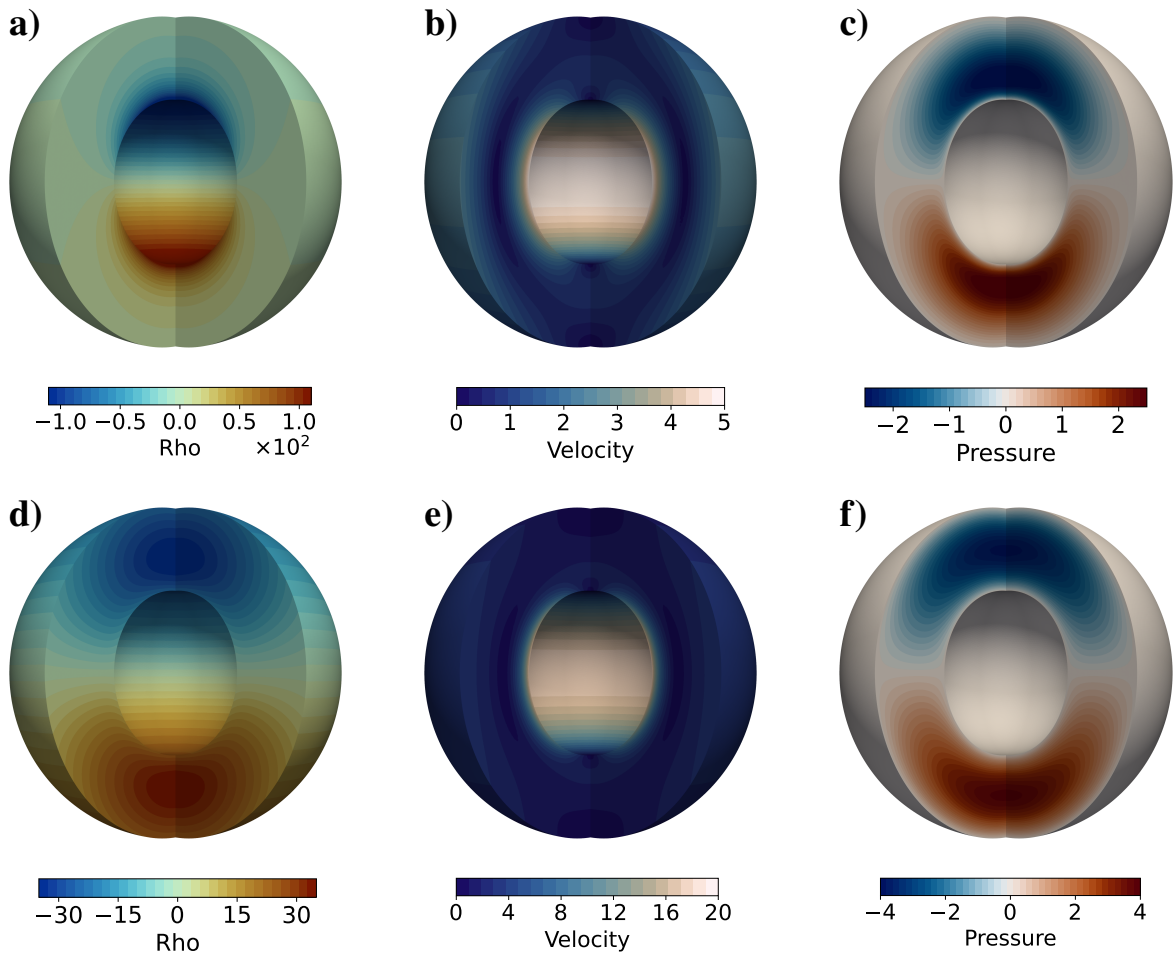

Figure 1. Analytical density, velocity magnitude, and pressure fields for the spherical Thieulot benchmark. The upper row shows the $m = -1$ case, and the lower row shows the $m = 3$ case.

Figure 1 illustrates the analytical fields used for verification. Both cases satisfy the no-normal-flow condition on the inner and outer boundaries while retaining non-zero tangential velocity. The $m = 3$ case uses the same shell geometry but introduces stronger radial viscosity variation, making pressure and stress recovery more demanding than in the constant-viscosity case.

3 Numerical setup

The benchmark is implemented in UW3 using the finite-element method. The reported calculations use a P_2P_1 Taylor–Hood element pair, with quadratic velocity and linear pressure. This pair is expected to give third-order velocity convergence and second-order pressure convergence in volume L_2 norms for sufficiently smooth Stokes solutions (Girault and Raviart, 1986; Boffi et al., 2013). Mesh resolution is reported using the nominal cell size $h = 1/8, \ldots, 1/128$. The analytical velocity is imposed on both spherical boundaries, and the pressure nullspace is handled by comparing mean-adjusted numerical and analytical pressure fields.

4 Error metrics

For a numerical quantity q_h and analytical field q^*, the relative volume error is

$$E_{L_2}^{\mathrm{rel}}(q) = \left(\frac{\int_\Omega |q_h - q^*|^2 \, d\Omega}{\int_\Omega |q^*|^2 \, d\Omega} \right)^{1/2}. \tag{10}$$

This metric is used for velocity and pressure. Boundary diagnostics are evaluated on the inner and outer spherical surfaces, denoted Γ_{inner} and Γ_{outer}, using

$$E_{L_2,\Gamma}^{\mathrm{rel}}(q) = \left(\frac{\int_\Gamma |q_h - q^*|^2 \, d\Gamma}{\int_\Gamma |q^*|^2 \, d\Gamma} \right)^{1/2}. \tag{11}$$

Absolute boundary pressure errors use the corresponding unnormalised measure

$$E_{L_2,\Gamma}(p) = \left(\int_\Gamma |p_h - p^*|^2 \, d\Gamma \right)^{1/2}. \tag{12}$$

Observed convergence rates between two consecutive mesh sizes h_1 and h_2 are computed as

$$r = \frac{\log(E_{h_1}/E_{h_2})}{\log(h_1/h_2)}. \tag{13}$$

5 Results

5.1 Velocity and pressure

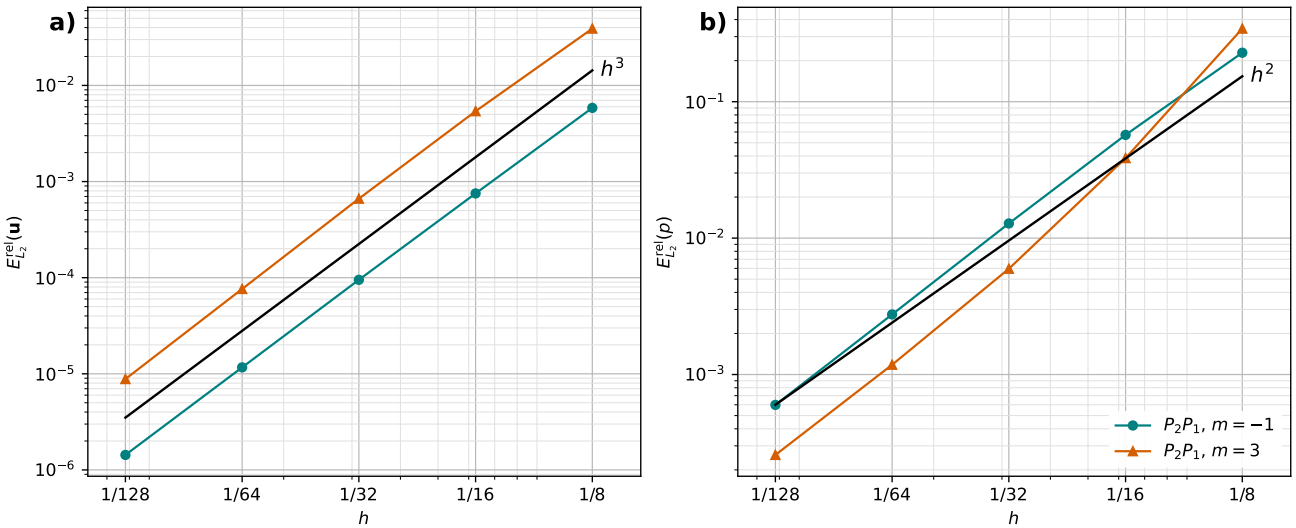

Figure 2. Convergence of the volume L_2-norm velocity and pressure errors for the P_2P_1 discretisation.

Table 1. Spherical Thieulot benchmark convergence for $m = -1$ and $m = 3$.

h	$m = -1$				$m = 3$			
	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	velocity rate	$E^{\mathrm{rel}}_{L_2}(p)$	pressure rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	velocity rate	$E^{\mathrm{rel}}_{L_2}(p)$	pressure rate
$1/8$	5.842e-03	–	2.287e-01	–	3.899e-02	–	3.433e-01	–
$1/16$	7.521e-04	2.96	5.709e-02	2.00	5.391e-03	2.85	3.858e-02	3.15
$1/32$	9.500e-05	2.98	1.281e-02	2.16	6.630e-04	3.02	5.959e-03	2.69
$1/64$	1.166e-05	3.03	2.757e-03	2.22	7.628e-05	3.12	1.179e-03	2.34
$1/128$	1.432e-06	3.03	5.986e-04	2.20	8.816e-06	3.11	2.573e-04	2.20

The velocity and pressure errors are shown in Fig. 2 and Table 1. The velocity error approaches $O(h^3)$ for both $m = -1$ and $m = 3$. The pressure error approaches $O(h^2)$ at the finest resolutions, with more irregular coarse-grid rates in the variable-viscosity case. This behaviour is consistent with the expected Taylor–Hood convergence hierarchy and with the convergence reported by Thieulot (2017).

5.2 Boundary stress and pressure

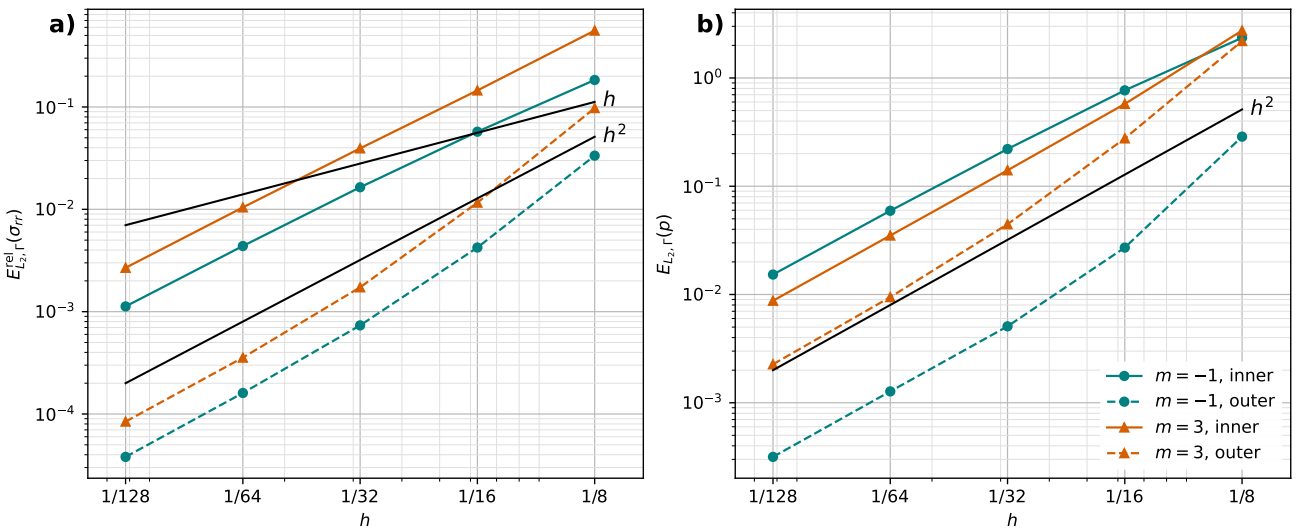

Figure 3. Convergence of boundary radial normal-stress and pressure errors. Solid lines denote the inner boundary and dashed lines denote the outer boundary.

Table 2. Spherical Thieulot benchmark convergence for boundary σ_{rr} errors.

h	$m=-1$				$m=3$			
	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{inner}}}(\sigma_{rr})$	inner rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{outer}}}(\sigma_{rr})$	outer rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{inner}}}(\sigma_{rr})$	inner rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{outer}}}(\sigma_{rr})$	outer rate
$1/8$	1.836e-01	–	3.343e-02	–	5.578e-01	–	9.755e-02	–
$1/16$	5.732e-02	1.68	4.233e-03	2.98	1.450e-01	1.94	1.162e-02	3.07
$1/32$	1.644e-02	1.80	7.347e-04	2.53	3.941e-02	1.88	1.730e-03	2.75
$1/64$	4.382e-03	1.91	1.605e-04	2.19	1.044e-02	1.92	3.556e-04	2.28
$1/128$	1.126e-03	1.96	3.811e-05	2.07	2.684e-03	1.96	8.435e-05	2.08

Table 3. Spherical Thieulot benchmark convergence for absolute boundary pressure errors.

h	$m=-1$				$m=3$			
	$E_{L_2,\Gamma_{\mathrm{inner}}}(p)$	inner rate	$E_{L_2,\Gamma_{\mathrm{outer}}}(p)$	outer rate	$E_{L_2,\Gamma_{\mathrm{inner}}}(p)$	inner rate	$E_{L_2,\Gamma_{\mathrm{outer}}}(p)$	outer rate
$1/8$	2.351e+00	–	2.870e-01	–	2.739e+00	–	2.192e+00	–
$1/16$	7.683e-01	1.61	2.712e-02	3.40	5.756e-01	2.25	2.766e-01	2.99
$1/32$	2.206e-01	1.80	5.082e-03	2.42	1.403e-01	2.04	4.455e-02	2.63
$1/64$	5.933e-02	1.89	1.273e-03	2.00	3.505e-02	2.00	9.456e-03	2.24
$1/128$	1.528e-02	1.96	3.162e-04	2.01	8.757e-03	2.00	2.267e-03	2.06

Although Thieulot (2017) derive the full stress tensor, the published benchmark focuses on velocity and pressure convergence. Here the radial normal stress, σ_{rr}, is included as an additional diagnostic because normal stress on spherical boundaries is important for dynamic-topography and surface loading calculations. Figure 3 and Table 2 show that σ_{rr} errors approach approximately second-order convergence on both shell boundaries for both viscosity cases. The boundary pressure errors show the same asymptotic trend.

6 Discussion

The results verify three aspects of the UW3 spherical-shell Stokes implementation. First, the P_2P_1 discretisation recovers the theoretical smooth-solution hierarchy: velocity converges one order faster than pressure in the L_2 norm. Second, the same rates are obtained for both constant and radially variable viscosity, although the $m = 3$ case has larger error constants and stronger pre-asymptotic pressure behaviour. Third, the boundary radial normal-stress diagnostic converges at roughly second order, consistent with related spherical-shell stress verification studies (Kramer et al., 2021).

The boundary error magnitudes differ between the inner and outer surfaces. This does not indicate a failure of the benchmark. The relative norms are normalised separately on two surfaces with different radii, different analytical stress amplitudes, different curvature, and different local combinations of pressure and deviatoric-stress error. The verification criterion is therefore the asymptotic slope, not equality of the error constants.

The benchmark is deliberately compact but still probes several practical issues for spherical geodynamics solvers: curved boundary representation, pressure normalisation, radial viscosity variation, tangential velocity boundary conditions, and stress recovery on spherical surfaces. These properties make it a useful regression test alongside annulus and Cartesian Stokes benchmarks.

7 Conclusions

We have presented a Copernicus-style verification article for the UW3 implementation of the Thieulot spherical-shell Stokes benchmark. The benchmark uses analytical velocity, pressure, density, viscosity, and stress fields in a curved three-dimensional shell. For the P_2P_1 Taylor–Hood discretisation, UW3 recovers approximately $O(h^3)$ velocity convergence and $O(h^2)$ pressure convergence for both the $m = -1$ and $m = 3$ cases. Boundary radial normal-stress and boundary pressure diagnostics also approach second-order convergence at the finest resolutions. These results support the use of the spherical Thieulot benchmark as a production verification case for UW3 velocity, pressure, and boundary-stress post-processing.

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